Predicting the Eurovision final from country pairs and Google searches

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Abstract

We predict the points every country gives every other country in the Eurovision final, using only what is known the day before: earlier votes between the two countries, Google searches for the entry in the voting country, distance, language and the running order. Summed per entry, the predictions give a scoreboard. Trained on 2008 to 2019 and chosen on 2021 to 2023, gradient boosting puts the real winner of the test finals 2024, 2025 and 2026 in first, second and fourth place. With three test finals and a measure we settled on after the first results, this is a hint and not proof. The whole scoreboard stays rough: 6.29 places off on average against 8.43 for a random order. Simple linear models do slightly better there. Earlier votes and searches carry the prediction, while the lyrics and the age and sex of the artist do not help.

1 Question and answer

In the Eurovision final 25 to 27 countries compete and every country taking part that year votes. Each vote gives 12, 10 and 8 down to 1 points to ten other countries, never to itself. Up to 2015 a country casts one vote. Since 2016 its jury and its public vote separately and both are published (Adamska and Reiss, 2025). We ask how well the scoreboard of the final can be predicted from data that exists the day before, above all its winner. Our canvas named two groups who would want to know: people who trade on prediction markets and the contestants themselves.

The answer has two sides. The model we chose on the validation finals puts the real winner of 2024, 2025 and 2026 in first, second and fourth place. On the whole scoreboard it is 6.29 places off on average against 8.43 for a random order. The simple linear models do slightly better than any tree.

2 Data

A row is one directed country pair in one final: country A votes for the entry of country B. Greece gave Cyprus 10 jury points and 12 public points in 2024, so that row has the target 11. Before that final Greece had given Cyprus 10.4 points on average in 27 earlier finals. We predict rows and not entries because the 18 finals from 2008 to 2026 (2020 was cancelled) have only 462 entries, too few for a dozen features, while they give 18 069 pair rows. Features such as distance or earlier votes also only exist for a pair. Up to 2015 the target is the single vote of a country. From 2016 it is half of jury plus public, so every year has one 0 to 12 scale and the sum per entry is half of its official points. Since 2023 viewers in countries without an entry vote together as "Rest of the World", a public vote without a jury that we halve like any other. Half of the pair rows are zero (50.3 percent), because only ten countries score per vote.

Table 1 lists the sources. Per pair we use the logarithm of the distance, a shared border, two measures of shared language, three of colonial ties plus the mean points and number of earlier meetings of this directed pair in all finals since 1957. Per entry we use the running order, the age and sex of solo artists as well as mood, theme and repetition of the lyrics. The lyrics were labelled by a language model through an OpenAI compatible interface at temperature 0, with fixed answer categories. The configured model is gpt-oss-120b on the e-INFRA CZ service, the output file does not record which model ran.

Google Trends gives the search interest in each voting country for each entry, for web search and YouTube, over the 30 days up to the day before the final. Trends does not report search volumes. It scales the country with the most interest to 100 and shares that scale between all terms of one query, so every entry is queried alone (Google). A value therefore says how much a country searched for an entry compared with the country that searched most. A teammate suspected that time zones could pull the final itself into the window. The query uses UTC and ends the day before, the final starts at about 19:00 UTC, so even a shifted day boundary ends the window at least seven hours before the show. We checked the stored windows for 2009, 2015, 2017 and 2024 to 2026 and the dates of all 17 finals with search data (2010 has none). The same script queried the years before 2016. The search data are weak before 2016: both values are zero in 98.7 percent of the rows in 2008, 72.6 percent in 2016 and 38.0 percent in 2024. 2010 has neither searches nor lyrics. Age and sex are missing for a third of the entries, mostly bands.

The pipeline is written in Python with pandas and stores its tables as Parquet. One command (`uv run esc all`) downloads, cleans and joins all sources on Linux and Windows. Network access happens only in the download step, every response is cached and every source is listed in `sources.yml`, with a checksum wherever the source has a fixed

Table 1: Sources. Every download is listed with address, date and checksum in `sources.yml`.

Source	What we take	Licence
Spijkervet (2023)	results and every vote up to 2023	CC BY 4.0
eurovisionworld.com	votes 2024 to 2026, from the data file its result pages load	none stated
CEPII GeoDist (Mayer and Zignago, 2011)	distance, border, language, colonial ties	Etalab 2.0
Google Trends, via pytrends	search interest per entry and voting country	Google terms, cited
Zviazdou (2026)	lyrics labels, trends 2008 to 2015	GPL-3.0
Wikidata	birth date and sex of solo artists	CC0
own tables	country codes, voting rules, final dates	CC0

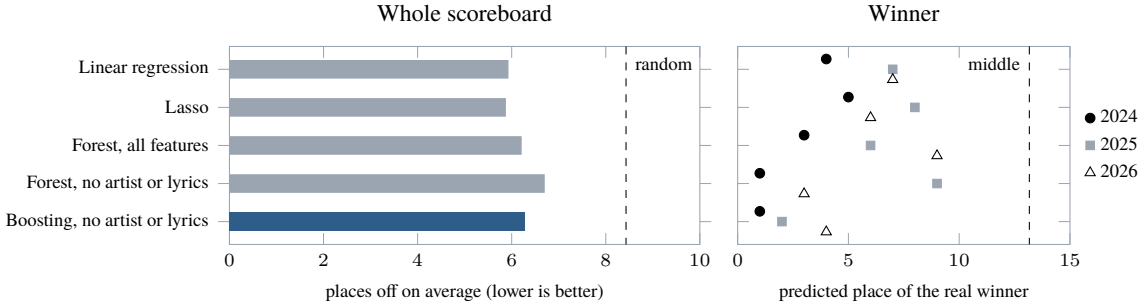


Figure 1: Test finals 2024 to 2026 for five of the seven models. The blue model was chosen on the validation finals. On the whole scoreboard all models are close and the simple linear ones do slightly better, while the chosen model puts the real winner highest. The right panel has the same rows as the left. Dashed: a random order (left, mean of 1000 draws) and the middle of the field (right).

release. The tests run offline. For the prediction a teammate extended the pipeline table to 2008, added age and sex and saved it as one CSV file, so the whole group could work without running the pipeline. For 2016 to 2026 it matches the pipeline table, which keeps jury and public as separate rows, in all 19 800 of these rows on points, earlier votes, distance and running order. The prediction notebook reads this file. In 31 percent of these rows the YouTube value differs, because the CSV file uses a later query of the same searches (team repository, commit e407c54). The code for the years before 2016 is not in the repository. A second notebook of the group plots what winners have in common (age, sex, running order, language, lyrics, searches) and fed the slides. The only people in the data are the artists, who are public figures.

3 Method

Nothing goes into the model that is measured after the vote. That rules out three things. The semi-final points are published after the final. The points other countries gave the same entry are the result itself. YouTube view counts are only available as today's count, which includes every view the winner got for winning. Adamska and Reiss (2025) point out this leak for their own view counts.

The finals are split by year and never at random, since the rows of one final depend on each other: training 2008 to 2019, validation 2021 to 2023, test 2024 to 2026. Missing values are filled with the median of the training years. We fit linear regression and lasso on standardised features (the lasso penalty by cross validation over groups of years), a random forest with 24 settings and gradient boosting with 8, all with scikit-learn (Pedregosa et al., 2011). The model with the best validation R^2 is chosen: gradient boosting without the artist and lyric features, learning rate 0.03, depth 3, at least 200 rows per leaf, 300 rounds. The test numbers of the other models were computed after this choice and appear only in Figure 1.

Predicted points are summed per entry and sorted into a scoreboard. We report the rank error, the mean distance between predicted and real place (Adamska and Reiss, 2025), the predicted place of the real winner and R^2 on the rows. For the rank error a random order is drawn 1000 times per final. For the winner a random order gives the middle of the field on average. Two checks follow. The forest with all features is refitted without one group of features at a time on the validation finals. And every final from 2016 is predicted once by the chosen model trained only on the finals before it. The slides of 15 October left out one year at a time and also trained on later years, whose earlier votes already contain the year being predicted. Training only on earlier years gives almost the same places for 2021 to 2026.

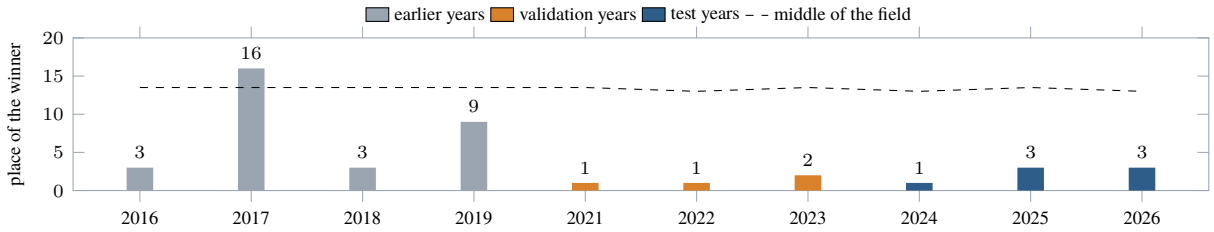


Figure 2: Every final from 2016 predicted once by the chosen model, trained only on the finals before it. The settings come from the validation years 2021 to 2023 (orange), so only 2024 to 2026 are fully independent.

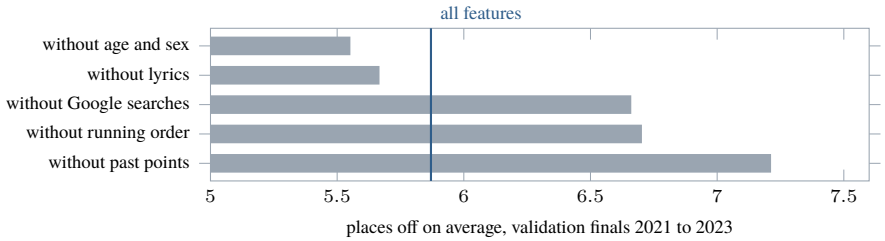


Figure 3: The forest with all features (blue line) against the same forest without one group of features. A bar ending right of the line means the group helped. This check led to the chosen model, which drops age, sex and lyrics.

4 Results

On the validation finals the chosen boosting reaches R^2 0.213 and is 5.53 places off, the forest with all features 0.179 and 5.87, lasso 0.125 and 6.96, linear regression 0.065 and 7.25, a random order 8.56. On the test finals (Figure 1) the boosting reaches R^2 0.152 and is 6.29 places off against 8.43 for a random order (95 percent range 7.18 to 9.61). It puts Switzerland, Austria and Bulgaria first, second and fourth. On the whole scoreboard lasso (5.88 places, R^2 0.184) and linear regression (5.93, 0.172) do slightly better than any tree. With three finals none of these differences is reliable. What the boosting does better is the top of the scoreboard. The slides of 15 October showed the test winners in places 1, 4 and 4, which are the numbers from leaving out single years.

Predicting every final from the earlier ones (Figure 2) puts the winner in the top three in eight of ten finals and first three times, at 6.29 places off against 8.53 for a random order. For 2024 to 2026 the places here are 1, 3 and 3 instead of 1, 2 and 4, because these models also learn from the finals 2021 to 2023. The finals from 2021 on look better than those before, partly because more training years and better search data were available, partly because the settings were chosen on 2021 to 2023. We cannot separate the two.

Figure 3 shows what carries the prediction. Without the earlier votes between the two countries the forest is 7.21 places off instead of 5.87, without running order 6.70 and without searches 6.66. Without lyrics (5.67) or age and sex (5.55) it gets better. The years 2008 to 2015 help despite their weak search data, 5.87 places with them and 7.17 without. How much of the earlier votes is neighbourhood is a question of its own that we leave aside here.

Our pipeline also predicts jury and public separately for 2016 to 2026, leaving out one year at a time and training on the rest. With pair features and searches the jury ranking is 6.40 places off and the public ranking 5.56, against 8.51 and 8.43 for a random order. Without searches it is 6.77 and 6.41. Searches help the public about twice as much as the jury. The two rankings correlate with Spearman 0.43 only. Computing the earlier votes from the training years alone changes these numbers by at most 0.13 places. The numbers are not directly comparable with Figure 1: the check differs and the feature set was chosen on the same years. For orientation, Adamska and Reiss (2025) also leave out one year at a time from 2008 to 2024 and reach 6.07 places in the final with audio, lyrics, YouTube views, running order and voting history, 6.05 with YouTube views alone and 7.59 with running order and voting history, which is closest to our features, against 8.55 for chance. We do not report win probabilities. In 2000 simulations per final drawn from the errors of the public model, eight entries got a win probability above 50 percent and only one of them won.

5 Limits

Three test finals are few and differences below one place are within chance. The winner's place became our main measure only after the first results were in. Before that we used the rank error. The test finals were looked at more than once: with the first forest, once to compare the forest with and without age and sex, with the chosen boosting and for the numbers of the other models in Figure 1. The boosting was chosen on the validation finals, but the test set was no longer untouched. On the whole scoreboard it is not the best model there. After we found that Rest of the World counted twice,

Table 2: What the canvas of 14 September planned and what we did.

Canvas	Done	Why
two public Eurovision datasets	Spijkervet up to 2023, eurovisionworld after	the second dataset has no final place, Spijkervet ends in 2023
one row per entry	one row per directed country pair	462 entries are too few, pair features need pairs
features from the songs	video dropped 10 Sep, audio 16 Sep	too much for the course; audio explains little in published work
bookmaker odds	dropped	no longer in the list of data fields of 17 Sep; nobody wrote down why
Wikipedia page views	Google Trends	Trends exists per country, so per pair
migration from Eurostat	dropped 20 Sep	we never confirmed the codes of the table
points normalised per year	half of jury plus public	one 0 to 12 scale in every year
missing values kept as missing	median of the training years	the models need complete rows
all songs since 1956	2008 to 2026 (2016 to 2026 until 15 Oct)	searches start in 2008, more years help

we fixed it and computed all numbers again without choosing anew. Search data before 2016 are mostly empty and age and sex are missing for bands. Averaging jury and public blurs two votes that agree only loosely. People on prediction markets would want win probabilities, which we do not trust enough to report. The prediction file is a snapshot whose pre-2016 code is not in the repository. Code and data live in a private repository of the group and the course staff can get access on request.

6 Communication and reflection

The spotlight presentation on 15 October explained the prediction to the target group. A blog post by Anna-Ilona follows with the submission. What our audience gets from it is a sense of how far a forecast from the day before can be trusted and which data matter for it.

Table 2 compares our canvas of 14 September with what we did. What worked was the pair as the unit, splitting by year, searches per country, a pipeline that runs on every machine of the group and one CSV file that let all three of us work without it. What did not work: lyrics add nothing, because a value that is the same for every voter of an entry lets the trees learn the entries by heart. The first Trends download asked for several terms at once and came back almost all zero, so each entry was queried again on 4 October. Scraping eurovisionworld with a browser failed on a teammate's laptop, which is why the pipeline has downloaded the data file behind the page since 12 October. The pipeline was hard for the others to follow, which is why the CSV file exists. Age and sex give no reliable gain and the win probabilities were badly calibrated.

We met in the workshops and wrote in a group chat. The TA approved the idea on 10 September. Data sit in one repository and the pipeline in another that reads the data at a fixed commit. Changes went in as pull requests. On 13 October it was unclear for a day whether the prediction or the neighbourhood question was our topic. On 14 October we settled on the prediction and split the work. Looking back we would decide on the measure before seeing any results, since the whole scoreboard and the winner tell different stories. We would keep the test finals closed until the end, share one simple file early, query Trends one entry at a time from the start and write the main question down in the first week. Our final slide named the next steps we see for the prediction: a neural network, the video clips themselves, better lyric sentiment and more search sources. To these we add cutting the searches two days before the final as Anna-Ilona suggested, a model that learns rankings directly, uncertainty by resampling years and moving the pre-2016 extension into the pipeline. Yauheni asked for a window that ends after the second semi-final, which the current one already does.

We learned that a random split leaks when rows belong together and that a measurement taken after the event is no forecast. A source has to be understood before it is queried, as Trends showed with its shared scale. Features that are the same for a whole entry tempt trees to memorise. Reproducible means that it runs on every machine in the group and not only on the one where the code was written. And the chosen measure decides how good a result looks: 6.29 places off and a winner in first, second and fourth place describe the same model.

7 Who did what

Tim built the pipeline with the jury and public models, queried the Trends data again per entry, extended the prediction notebook with the checks and the boosting and wrote this report. Yauheni collected and labelled the lyrics, added the searches for 2008 to 2015 and the Wikidata fields, built the CSV file, wrote the first version of the prediction notebook and the notebook on winners and made the slides. Anna-Ilona ran the first Trends download, worked on the slides and

writes the blog post.

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Use of AI. Parts of this report were drafted with the help of an AI assistant (Claude Opus 5.5 by Anthropic) and revised by the authors.